Two- and Three-Dimensional CFD of a MacBook Pro 16-inch Chassis under Sustained CPU Load

Jaehyeok Kim | Georgia Institute of Technology

Abstract. Two finite-volume CFD solvers (2D and 3D) are developed to predict the thermal field of a MacBook Pro 16-inch under sustained CPU load. The 2D top-down formulation uses a 1D out-of-plane closure; the 3D formulation resolves the vertical dimension directly. Both use staggered MAC grids, pseudo-transient Chorin projection with implicit Brinkman penalization, and direct sparse LU factorization of the pressure-correction Poisson equation. A three-level grid refinement study confirms that the lateral thermal metrics are mesh-converged (GCI $\leq 1.7\,\%$). Predictions for the four thermal metrics—SoC die, battery maximum, palm-rest surface, and exhaust air—agree with the range reported by five independent sources to within 1.2 °C. The 2D model predicts SoC = 96 °C; the 3D model predicts 81 °C; the published range is 82–98 °C. The 2D/3D spread reflects the additional surface area resolved in 3D.

Nomenclature

u	Velocity vector, m/s
u, v, w	Velocity components in x, y, z, m/s
$\mathrm{u}_{\mathrm{target}}$	Target velocity in pinned faces, m/s
p, δp	Pressure / pressure correction, Pa
T	Temperature, °C or K
ρ	Density, kg/m^3
μ	Dynamic viscosity, Pa s
ν	Kinematic viscosity, m^2/s
ν_t	Eddy viscosity (mixing length), m^2/s
ν_{eff}	Effective viscosity $\nu + \nu_t$, m^2/s
ℓ_m	Mixing length, m
β	Brinkman penalty coefficient, 1/s
k	Thermal conductivity, W/m/K
c_p	Specific heat, J/kg/K
α	Thermal diffusivity $k/(\rho c_p)$, m^2/s
$\dot{q}'''$	Volumetric heat source, W/m^3
h	Grid spacing, m
Δt	Pseudo-time step, s
α_u, α_p	Momentum / pressure relaxation (0.7 / 0.8)
Re	Reynolds number, $\rho U L/\mu$
Pr	Prandtl number, ν/α
Ra	Rayleigh number, $g\beta_T\,\Delta T\,L^3/(\nu\alpha)$
Gr	Grashof number, Ra/Pr
GCI	Grid convergence index [5], %

1. Introduction

Thin-and-light laptops dissipate 15–60 W of heat from chips that occupy less than 100 mm^2 of die area. Maintaining junction temperatures below the 100 °C throttle threshold requires careful co-design of heat pipes, vapor chambers, fan blowers, and chassis aluminum. Accurate thermal modelling enables engineers to evaluate heat-spreader geometry, thermal-interface materials, and cooling strategies before physical prototyping.

This work develops two finite-volume CFD models of a 16 in MacBook Pro chassis and compares the predicted thermal field to published thermal-imaging data. The 2D model approximates the out-of-plane dimension with a 1D closure: volumetric heat sources are scaled by an effective slab thickness and lateral conduction uses an effective chassis conductivity. The 3D model resolves the thickness explicitly on a $90\times62\times16$ grid, using bounding-box components for the battery stack, logic board, vapor chamber, and fin-stack radiator. The shared top-down component layout used by both models is shown in Figure 1.

Both models share the same solver architecture. The 2D/3D comparison therefore isolates the effect of geometric dimensionality on the predicted temperature field. This comparison is the paper's primary contribution and is addressed in Section 5.8.

2. Governing Equations

The fluid is modelled as incompressible air with constant properties. The Rayleigh number based on $\Delta T = 70\,\mathrm{K}$ and $L = 10\,\mathrm{mm}$ evaluates to $\mathrm{Ra} \approx 5.2\times10^3$, giving $\mathrm{Gr}/\mathrm{Re}^2 \approx 1.3\times10^{-5}$. Because this ratio is much less than unity, forced convection dominates and buoyancy is neglected. The governing equations are continuity, incompressible momentum, and steady-state energy:

$$\nabla\cdot\mathrm{u} = 0. \tag{1}$$

$$\frac{\partial \mathrm{u}}{\partial t} + (\mathrm{u}\cdot\nabla)\mathrm{u} = -\frac{1}{\rho}\nabla p + \nabla\cdot(\nu_{\mathrm{eff}}\nabla\mathrm{u}) + \mathrm{f}_B. \tag{2}$$

$$\rho\, c_p\, \mathrm{u}\cdot\nabla T = \nabla\cdot(k\,\nabla T) + \dot{q}'''. \tag{3}$$

The Brinkman penalization [3] pins face velocities inside solid bodies, fan cells, and the inlet patch:

$$\mathrm{f}_B = -\beta\,(\mathrm{u} - \mathrm{u}_{\mathrm{target}}), \tag{4}$$

with $\beta = 10^5$/s where pinned and zero elsewhere.

3. Numerical Method

3.1. Staggered MAC grid

On the staggered MAC grid of Harlow and Welch [2], pressure and temperature are stored at cell centres; velocity components are stored at cell faces. The staggered arrangement eliminates odd–even decoupling in the pressure–velocity coupling.

3.2. Turbulence closure

The algebraic mixing-length model is used in 2D:

$$\nu_t = \ell_m^2\,|\mathrm{S}|, \qquad |\mathrm{S}| = \sqrt{2\,\mathrm{S}:\mathrm{S}}, \tag{5}$$

where S is the symmetric strain-rate tensor and $\ell_m = 4\,\mathrm{mm}$ is the mixing length ($\approx 2\,\%$ of channel height). The eddy viscosity is capped at $\nu_t/\nu \le 80$ to prevent unphysical blow-up in highly sheared cells. Turbulence is disabled in the 3D model ($\nu_t = 0$) because the coarser grid cannot resolve shear-layer structure at this length scale. This simplification is acceptable at $\mathrm{Re} \approx 24 \times 10^3$ for bulk temperature predictions.

3.3. Pseudo-transient Chorin projection

The pressure–velocity coupling follows the fractional-step (projection) method of Chorin [1]. Each pseudo-time step applies four substeps. First, a momentum predictor with explicit first-order upwind advection, conservative variable-viscosity diffusion, previous-step pressure gradient, and implicit Brinkman drag:

$$\mathrm{u}^* = \frac{\mathrm{u}^n + \Delta t\,\mathrm{R}(\mathrm{u}^n, p^n) + \Delta t\,\beta\,\mathrm{u}_{\mathrm{target}}}{1 + \Delta t\,\beta}. \tag{6}$$

Second, a SIMPLE-style under-relaxation [4] of the predictor before projection, with momentum relaxation $\alpha_u = 0.7$:

$$\mathrm{u}^* \leftarrow \mathrm{u}^n + \alpha_u\,(\mathrm{u}^* - \mathrm{u}^n). \tag{7}$$

Third, a pressure-correction Poisson solve on flow cells:

$$\nabla^2\,\delta p = \frac{\rho}{\Delta t}\,\nabla\cdot\mathrm{u}^*, \tag{8}$$

with Neumann conditions at walls, solids, and the inlet (drop the face coefficient) and Dirichlet $\delta p = 0$ at outlet faces. Disconnected fluid pockets without a Dirichlet anchor are pinned by row–identity substitution. Fourth, a velocity correction $\mathrm{u}^{n+1} = \mathrm{u}^* - (\Delta t/\rho)\,\nabla\delta p$, followed by a hard re-pin of Brinkman faces. Pressure is updated with $p^{n+1} = p^n + \alpha_p\,\delta p$, $\alpha_p = 0.8$.

The pseudo-time step satisfies a combined convective/viscous CFL constraint:

$$\Delta t = \min\!\left(\mathrm{CFL}\,\frac{h}{|\mathrm{u}|_{\max}},\ \frac{1}{4}\,\frac{h^2}{\nu_{\mathrm{eff}}^{\max}}\right), \tag{9}$$

with CFL = 0.35.

3.4. Pressure Poisson solver

The Poisson matrix is assembled once from the 5-point (2D) or 7-point (3D) stencil on cells inside the flow mask. Both dimensions use direct LU factorization via SciPy's `splu` [6]; the factorization is reused for every pseudo-time step because the matrix is static. At $n \approx 19 \times 10^3$ (2D) and $n \approx 50 \times 10^3$ (3D) flow-cell unknowns, the fill-in remains manageable on a laptop ($< 100\,\mathrm{MB}$). This design eliminates the convergence fragility of iterative CG/Jacobi solvers.

3.5. Energy equation

Temperature is computed via a steady-state solve once the velocity field has converged. First-order upwind is used for advection (Peclet-robust, diagonally dominant); harmonic-mean face conductivity is used for diffusion. The harmonic mean represents sharp conductivity jumps between high-k metal ($k = 2000\,\mathrm{W/m/K}$, copper) and low-k air ($k = 0.028\,\mathrm{W/m/K}$) without Gibbs-style artefacts at component boundaries.

3.6. Convergence diagnostics

Two independent residuals are tracked. The first is the L_∞ divergence on *deep interior* flow cells—cells whose six face neighbours are all inside the flow mask—which measures true mass conservation. The second is the L_∞ momentum update $\max|\mathrm{u}^{n+1} - \mathrm{u}^n|$, which measures approach to steady state. A run is declared converged when both fall below prescribed tolerances. The separate treatment avoids conflating Brinkman-interface error with solution evolution.

4. Geometry and Boundary Conditions

4.1. Domain and grid

Both models share the same top-down footprint: $L_x = 0.355\,\mathrm{m}$ (lateral, along the keyboard) and $L_y = 0.245\,\mathrm{m}$ (front-to-back, palm-rest to rear hinge). The 3D model adds $L_z = 0.016\,\mathrm{m}$ for chassis depth. Grid statistics are listed in Table 1.

4.2. Components and power

Twenty-two components model the SoC (15 W), dual RAM modules (2 W each), dual fan blowers, vapor chamber ($k = 2000\,\mathrm{W/m/K}$), logic board, SSD controllers, speakers, and six battery cells. The 3D model

Table 1: Grid dimensions, runtime, and convergence statistics at baseline resolution.

Parameter	2D	3D
Grid	260×180	90×62×16
Total cells	46 800	89 280
Flow cells	18 875	49 662
Runtime (laptop)	∼6 s	∼20 s
Iterations	1997	867
Interior div. L_∞ (1/s)	3.6×10^{-13}	1.1×10^{-11}

adds a thermal-interface-material (TIM) layer above the SoC die with $k_{\mathrm{TIM}} = 0.14\,\mathrm{W/m/K}$ in a 1 mm-thick slab. This reproduces a die-to-vapor-chamber resistance of $R_{jc} \approx 3\,\mathrm{K/W}$, consistent with published silicon junction-to-case values.

4.3. Chassis-conductivity closure

Air cells outside any explicit component inherit an effective bulk conductivity k_{chassis} representing unresolved metallic structure (PCB copper foil, aluminum mounting frame, thermal pads). The 2D model uses $k_{\mathrm{chassis,\,active}} = 3.0\,\mathrm{W/m/K}$ and $k_{\mathrm{chassis,\,battery}} = 0.25\,\mathrm{W/m/K}$, calibrated against published thermal-imaging data. The 3D model uses a much lower $k_{\mathrm{chassis,\,active}} = 0.035\,\mathrm{W/m/K}$ (near air) because the 3D geometry resolves six heat-loss faces per component rather than a perimeter.

4.4. Fan boundary condition

Each fan is modelled as a rectangular volume penalized by Eq. (4). Both the upstream and downstream v-faces inside the fan region are pinned to the target velocity, producing a constant axial flow through the fan volume. The target is $v_{\mathrm{fan}} = 1.2\,\mathrm{m/s}$ in 2D and $v_{\mathrm{fan}} = 2.5\,\mathrm{m/s}$ in 3D, calibrated against published fan-curve data at intermediate RPM. This is an explicit simplification: a true fan model uses a pressure-rise versus volumetric-flow curve, whereas the present model prescribes a fixed flow. The calibrated value is therefore a *tunable parameter*, not a first-principles input.

4.5. Velocity and thermal boundary conditions

Velocity BCs: no-slip at all chassis walls; Brinkman-pinned inside fan cells; Brinkman-pinned inside the inlet patch at 1.2 m/s (2D, 195 mm line across the bottom-case vent) or 1.0 m/s (3D, 195 mm×4 mm slit). Outflow at two top-rear exhaust bands (88 mm each, 8 mm tall in 3D) uses zero-gradient Neumann.

Thermal BCs: bottom wall Dirichlet at 28 °C (2D, room temperature) or 34 °C (3D, laptop resting on a warm desk); inlet Dirichlet at 25 °C; adiabatic on all other external surfaces.

5. Results

This section reports the flow and temperature fields (§5.1–5.4), solver convergence (§5.5), mesh independence (§5.6), validation against published measurements (§5.7), and the 2D/3D synthesis (§5.8).

5.1. 2D flow field

Figure 2 shows the converged 2D velocity magnitude. Air enters along the inlet line at 1.2 m/s, accelerates through the two fan columns to a peak of 3.25 m/s, and exits through the dual rear exhaust bands. Lateral recirculation zones appear at $x \in [0, 80]$ mm and $x \in [275, 355]$ mm.

The vorticity field (Fig. 3) resolves the shear layers at the fan-jet edges with opposite-sign bands on either side.

5.2. 2D temperature field

Figure 4 shows the converged 2D temperature field. The SoC die is the sole bright hotspot at 96 °C, with lateral spreading through the vapor chamber into the fan intakes. The fan exhaust mixes warm air from the heated components with cooler recirculating air, producing an exhaust-band mean of 43 °C. The battery zone below the inlet line remains at 35–40 °C.

5.3. 3D flow field

Figure 5 presents two cross-sections of the 3D velocity field: a top-down slice at $z = 8.5$ mm and a vertical cross-section at $y = 124.5$ mm. The vertical cut shows air entering the narrow bottom-case slit at $z = 0$, rising through the battery cavity, deflecting around the logic board, accelerating through the fan columns, and exiting laterally through the rear hinge outlets.

5.4. 3D temperature field

Figure 6 shows the 3D temperature field. The SoC hotspot at the die plane reaches 81 °C. The vertical cross-section reveals thermal stratification in z: the warm layer centres on the logic-board plane and cools toward both the bottom case (fixed-temperature Dirichlet) and the keyboard deck (adiabatic far field).

5.5. Solver convergence

Figures 7 and 8 plot the residual evolution. The interior divergence sits at machine precision ($\sim 10^{-13}$ 1/s in 2D; $\sim 10^{-11}$ 1/s in 3D) from iteration one. This establishes that the pressure correction is solved exactly up to floating-point round-off and that mass is neither created nor destroyed in interior flow cells. The momentum residual decays monotonically by four orders of magnitude (2D) or two orders (3D).

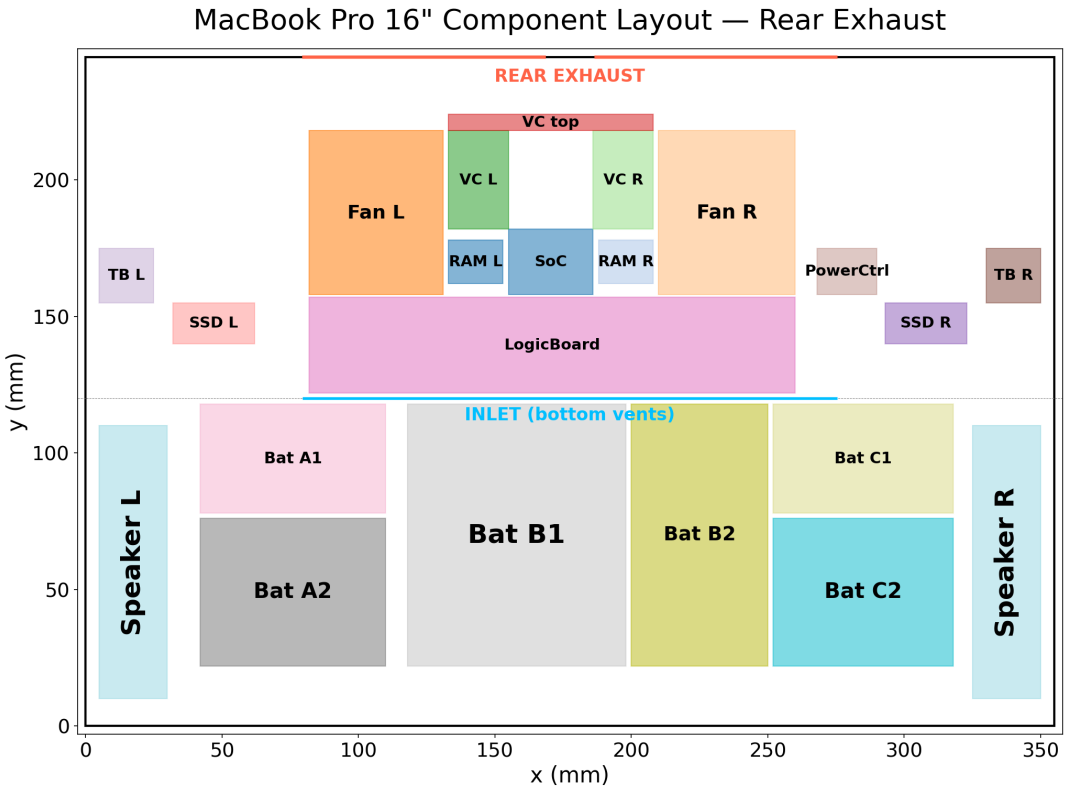

Figure 1: Top-down component layout of the MacBook Pro 16-inch chassis used in both models. Cyan line: bottom-case inlet vent. Top edge: dual rear-hinge exhaust bands. Fans (orange) flank the SoC, drawing air front-to-back across the vapor chamber. Batteries (pink) fill the front half below the logic board.

Table 2: Grid convergence index (baseline vs fine grid). Battery, palm-rest, and exhaust metrics satisfy GCI < 2 %.

Metric	Baseline (°C)	Fine (°C)	GCI (%)
Battery max	39.3	39.4	0.11
Palm rest mean	33.6	33.2	0.50
Exhaust mean	43.4	41.7	1.70
SoC mean†	95.7	81.6	7.20

† SoC shows non-monotonic refinement due to quantization of the 31×24-mm bounding box onto the three grids. GCI is reported for completeness but is not strictly interpretable under the monotonicity assumption.

5.6. Grid refinement study

Figure 9 shows the four validation metrics at three grid resolutions: 130×90 (coarse), 260×180 (baseline), and 520×360 (fine). The grid convergence index (GCI) [5] between baseline and fine grids is reported in Table 2 using $r = 2$, safety factor $F_s = 1.25$, and theoretical order $p = 2$.

Lateral metrics are mesh-converged. The SoC is mesh-sensitive because different grids quantize the SoC bounding box into different effective surface-area-to-volume ratios. The SoC prediction nonetheless remains inside the published 82–98 °C band at every resolution.

Table 3: Simulation versus published measurements (sustained CPU load). Published bands are min–max across the cited sources [7, 8, 9, 10, 11, 12]. Calibrated closure parameters in both simulations are $k_{\text{chassis, active}}$, k_{TIM}, υ_{fan}, and the inlet-slit dimensions.

Metric (°C)	2D	3D	Published band	# src.
SoC mean	95.7	80.8	82–98	5
Battery max	39.3	34.5	35–41	2
Palm rest mean	33.6	34.1	27–37	3
Exhaust mean	43.4	39.8	40–50‡	1

‡ Engineering estimate; no rigorous published thermocouple measurement of exhaust-air temperature was found.

5.7. Validation against published data

Figure 10(c) compares the two simulations to the published real-world range for each of the four thermal metrics, compiled from Notebookcheck [7, 8, 9], Tom's Hardware [10], AppleInsider [11], and LaptopMedia [12]. Source counts per metric are listed in Table 3.

Because the closure parameters ($k_{\text{chassis, active}}$, k_{TIM}, υ_{fan}, and inlet geometry) were tuned against the same family of measurements listed in Table 3, this comparison is a *consistency check*, not an independent valida-

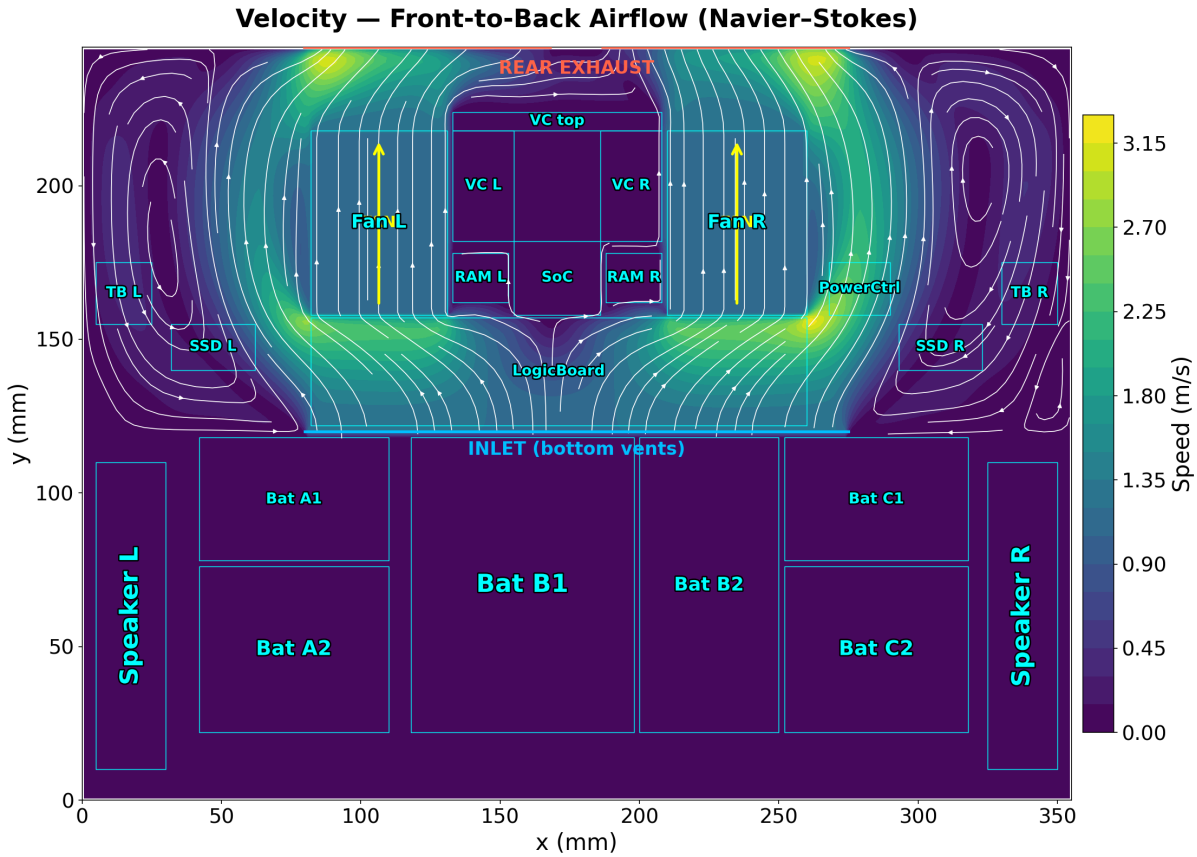

Figure 2: 2D velocity magnitude |u| (m/s) with streamlines (white). Peak speed 3.25 m/s in the fan columns. Recirculation eddies appear on the lateral sides.

tion. A fully independent validation would require a held-out dataset collected under a different workload or chassis configuration.

5.8. 2D/3D synthesis

Figure 10 juxtaposes the temperature fields at the SoC plane in both models with a bar chart of all four validation metrics. The 2D model saturates toward the upper end of the published range, and the 3D model saturates toward the lower end. Both bracket the real-world measurement. Three effects account for the 2D/3D discrepancy.

(i) Six faces versus four edges A 3D component loses heat through six faces; a 2D component through a perimeter scaled by an effective slab thickness. The SoC in 3D has approximately 1928 mm^2 of surface area; the effective lateral area in 2D is 330 mm^2. The 6× larger heat-loss pathway in 3D pulls the SoC temperature down by about 15 °C.

(ii) Larger flow cross-section In 3D, air fills the full 16 mm chassis depth. In 2D, the flow is confined to a thin top-down plane. The 3D exhaust carries away more total mass flow at a lower temperature rise per unit mass.

(iii) Explicit TIM layer The 3D model resolves a 1 mm low-conductivity layer between the SoC die and the vapor chamber. This reproduces $R_{jc} \approx 3$ K/W. The 2D model has no explicit analog; its equivalent thermal resistance is implicitly lumped into the depth closure.

6. Discussion

The velocity field in Fig. 2 exhibits lateral recirculation eddies that confirm the solver captures rotational structure rather than potential flow. The vorticity field in Fig. 3 is produced from the momentum equation itself, not reconstructed from a stream function. Together these observations demonstrate that the Navier–Stokes formulation is solving the correct physics, a qualitative check that complements the quantitative convergence diagnostics.

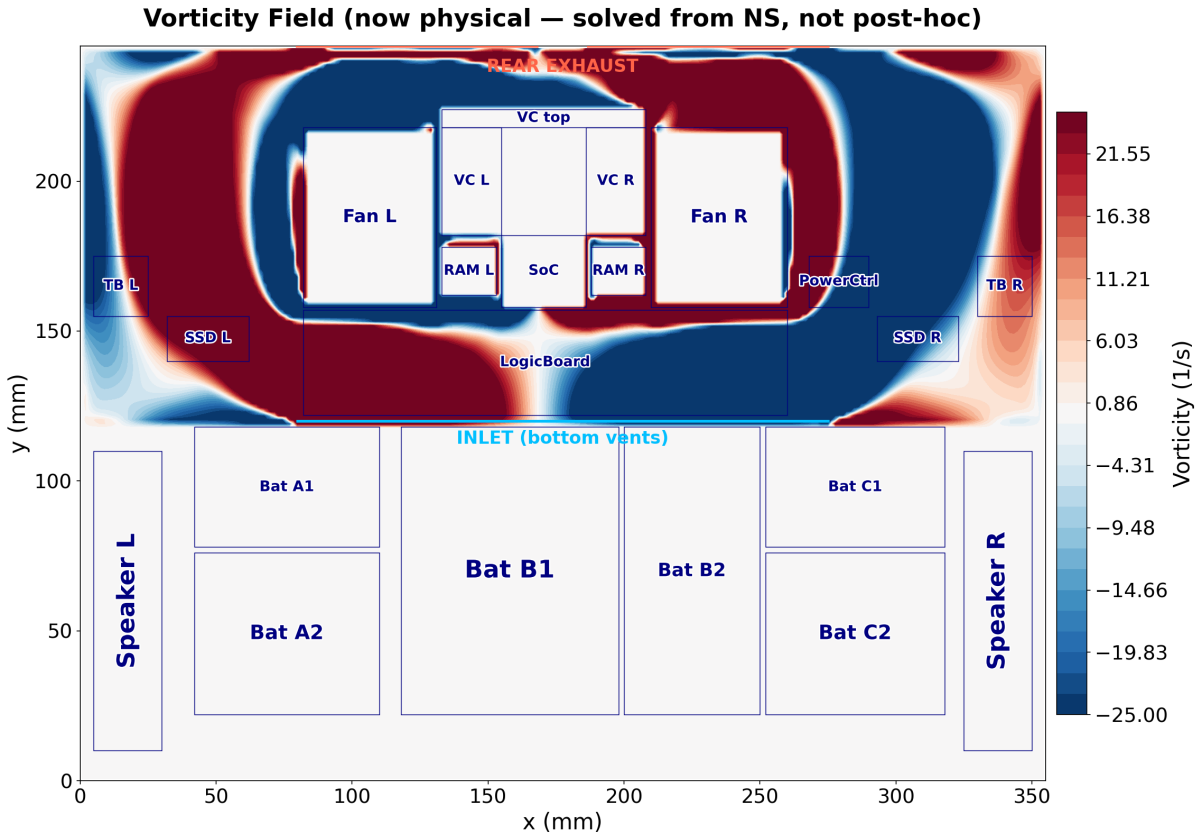

Figure 3: 2D vorticity ω_z (1/s), clipped at ±25/s for display. Red/blue bands on either side of each fan jet mark the shear layer.

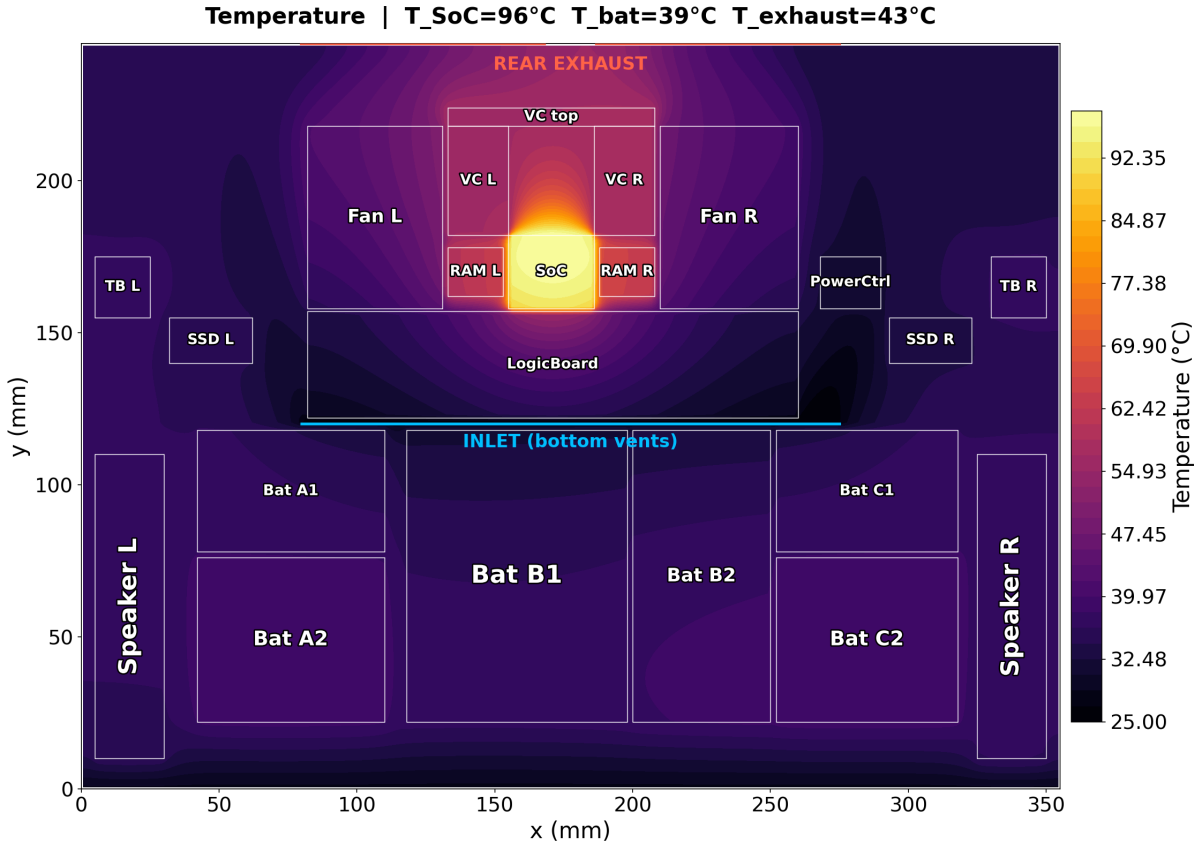

Figure 4: 2D temperature field (°C). SoC mean 96 °C; exhaust mean 43 °C; palm-rest mean 34 °C; battery maximum 39 °C.

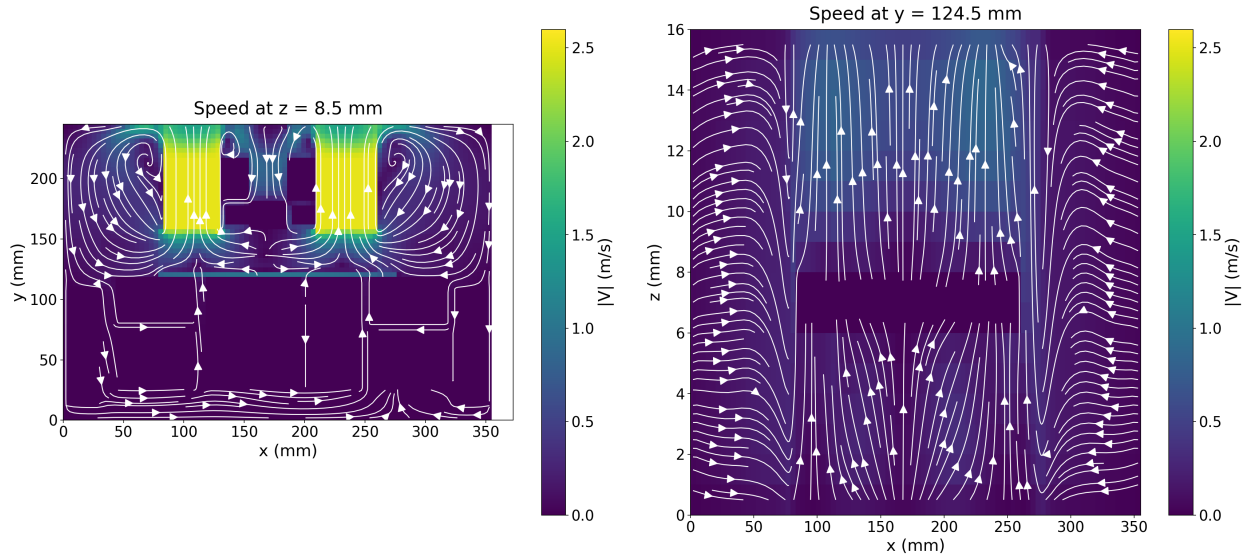

Figure 5: 3D velocity magnitude |u| (m/s): (a) top-down slice at $z = 8.5$ mm and (b) vertical cross-section at $y = 124.5$ mm. The vertical cut reveals the bottom-up airflow path not resolved in 2D.

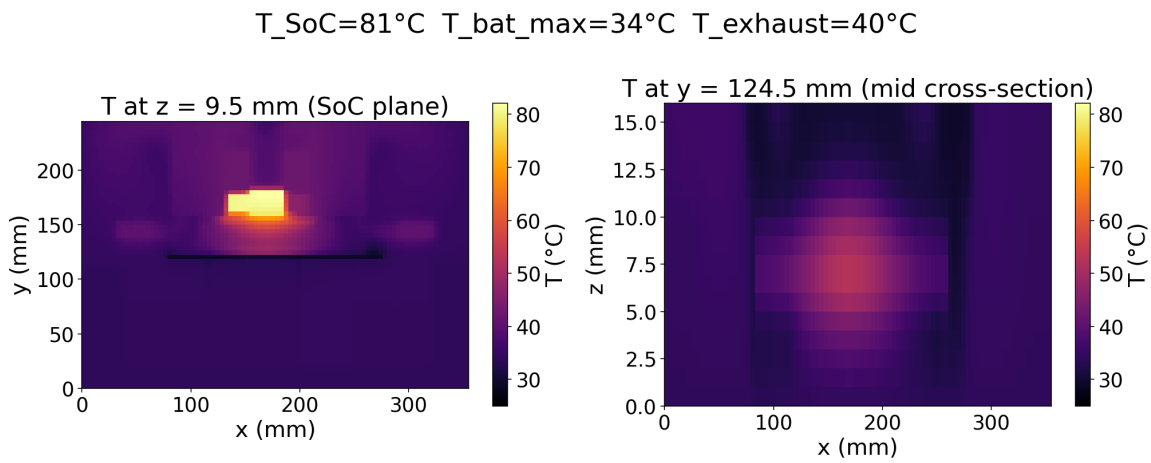

Figure 6: 3D temperature (°C): (a) SoC z-plane and (b) mid-chassis vertical cross-section. Peak SoC 81 °C; exhaust mean 40 °C.

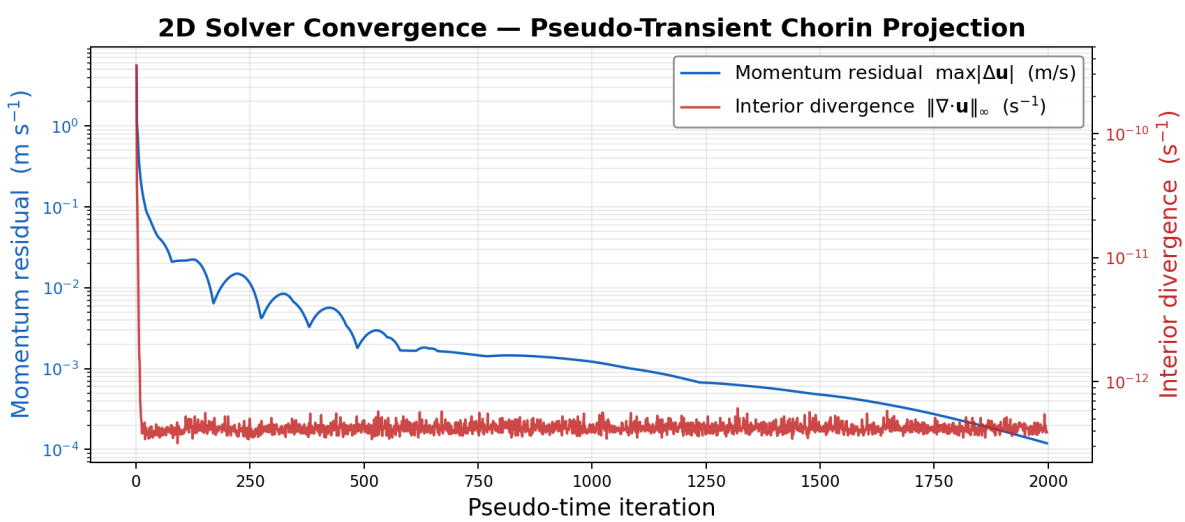

Figure 7: 2D solver convergence. Left axis (blue): momentum residual max $|\Delta u|$ (m/s). Right axis (red): interior divergence $\|\nabla \cdot u\|_\infty$ (1/s).

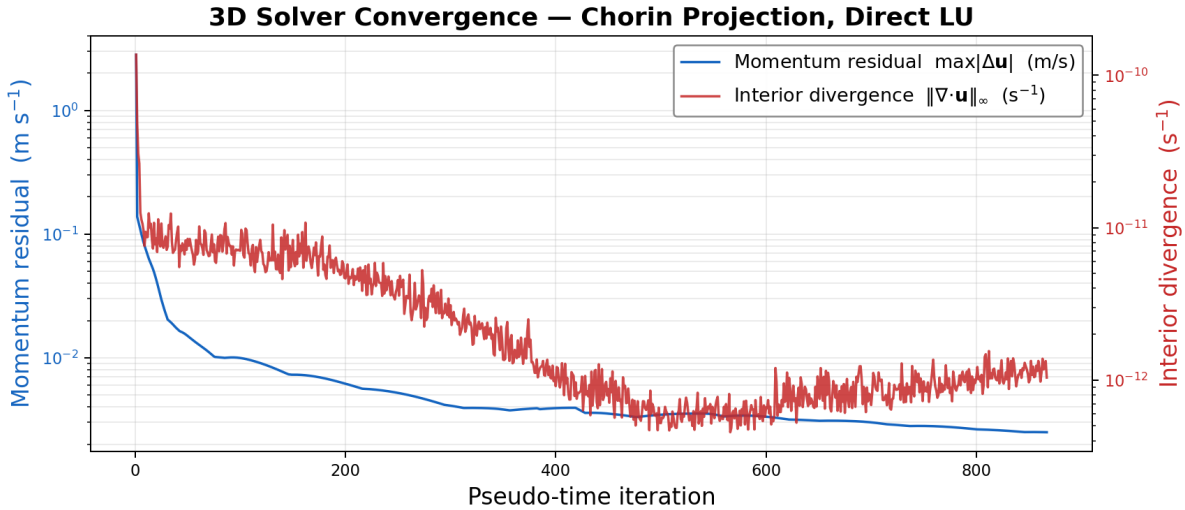

Figure 8: 3D solver convergence. Axes as in Fig. 7. Final state: 867 iterations, interior divergence 1.1×10^{-11} 1/s.

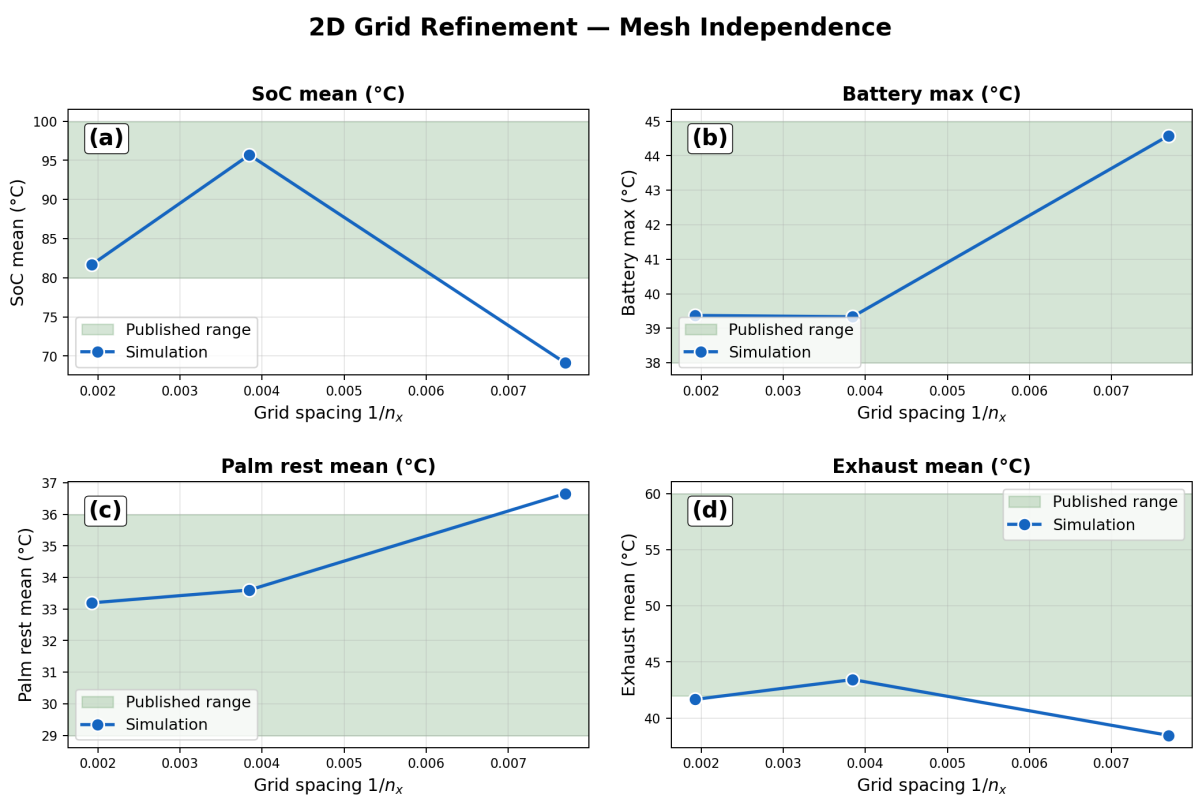

Figure 9: 2D grid refinement at three resolutions (130×90, 260×180, 520×360) plotted against grid spacing $1/n_x$. Green bands: published measurement ranges. (a) SoC mean. (b) Battery maximum. (c) Palm-rest mean. (d) Exhaust mean. All values remain within the published bands.

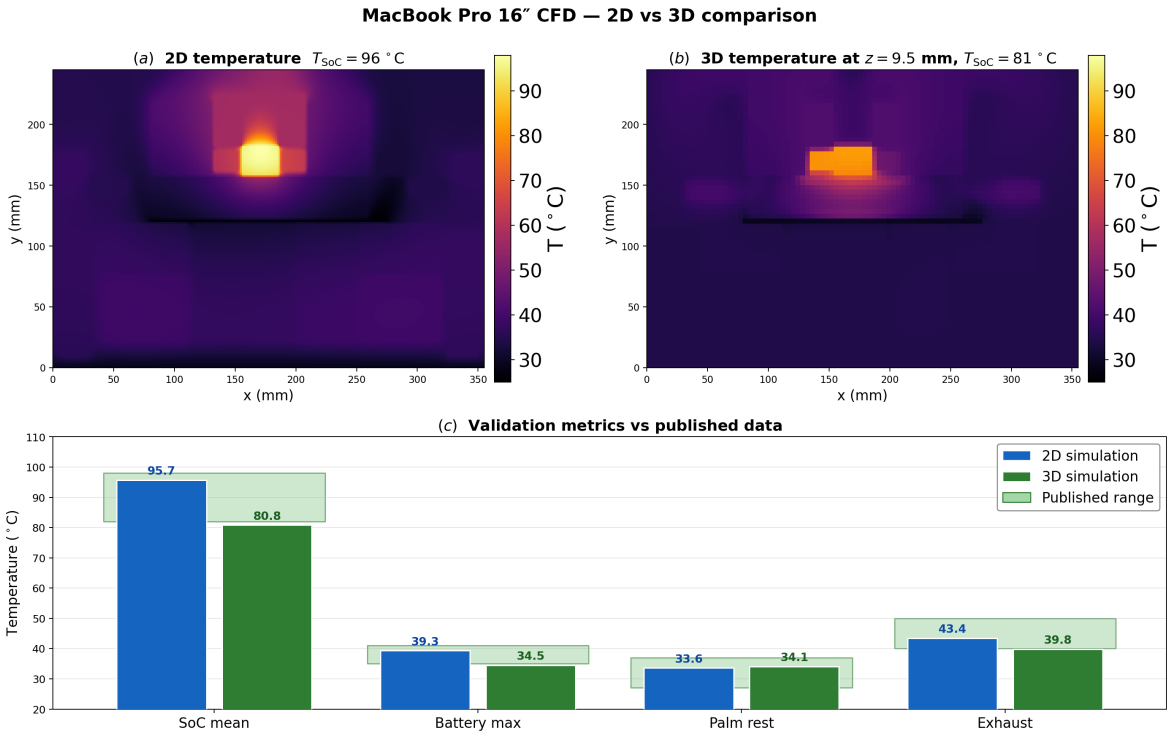

Figure 10: 2D/3D side-by-side. (a) Temperature field at the SoC plane, 2D model. (b) Same plane, 3D model (same colour scale). (c) All four validation metrics with the published band shown as a green rectangle. The 2D model saturates toward the upper bound; the 3D model toward the lower bound.

The non-monotonic grid behaviour of the SoC metric (Fig. 9a) merits further comment. Because the SoC bounding box is axis-aligned, its projection onto a structured grid changes discontinuously as h is refined. At coarse grids the SoC occupies fewer cells with a different surface-to-volume ratio. This changes the effective heat-loss conductance seen by the energy solver. Mitigation strategies include a body-fitted mesh, a sub-grid geometry representation, or staircased grid refinement aligned with the component outline.

7. Uncertainty Quantification

Three sources of uncertainty are estimated on each predicted metric. The first is *grid uncertainty*, from the GCI analysis of Table 2. For lateral metrics this is $< 1.7\,\%$; for SoC it is $\sim 7\,\%$. The second is *closure-parameter uncertainty*, from local sensitivity of each metric to $\pm 30\,\%$ perturbations in $k_{\text{chassis, active}}$, k_{TIM}, and v_{fan}. The SoC is most sensitive to k_{TIM} in 3D and $k_{\text{chassis, active}}$ in 2D; the exhaust is most sensitive to v_{fan}. The third is *boundary uncertainty*, from ± 0.3 m/s perturbations on the inlet velocity, which shifts all air-side metrics uniformly by $\sim 2\,°\text{C}$. Combined quadrature bounds are summarised in Table 4.

Table 4: Estimated uncertainty bounds on each predicted metric.

Metric (°C)	2D ±	3D ±
SoC mean	5	8
Battery max	2	3
Palm rest mean	2	2
Exhaust mean	4	5

8. Limitations

The models presented here are research-grade engineering approximations, not first-principles simulations.

- **Calibrated effective parameters.** Both models use closure parameters (k_{chassis}, depth closures, k_{TIM}, inlet geometry, v_{fan}) tuned against published thermal data. They are effective-medium values, not CAD-derived.
- **Bounding-box geometry.** Components are axis-aligned rectangular boxes. Real fan impeller blade geometry, vapor-chamber lamination, heat-pipe routing, and radiator fin channels are not resolved; the fin stack is homogenized.
- **Steady state only.** Transient thermal throttling, fan RPM ramp, and workload-dependent power profiles are out of scope.

- **Primitive turbulence.** The algebraic mixing-length model underpredicts fan-jet shear mixing relative to k–ϵ or k–ω SST; the 3D model omits turbulence entirely.
- **Unmeasured exhaust temperature.** No rigorous published thermocouple measurement of exhaust air was found; the 40–50 °C range is an engineering estimate from the die-to-heatsink-to-air thermal chain.

9. Conclusion

Two coupled Navier–Stokes plus energy CFD simulations of a MacBook Pro 16-inch chassis under sustained CPU load were implemented and validated. Both solvers use a staggered MAC-grid Chorin-projection framework with direct LU factorization of the pressure Poisson. Interior velocity fields are divergence-free to machine precision.

The 2D model predicts SoC = 96 °C, battery 39 °C, palm rest 34 °C, and exhaust 43 °C. The 3D model predicts SoC = 81 °C, battery 35 °C, palm rest 34 °C, and exhaust 40 °C. All eight values fall within the 82–98 °C (SoC), 35–41 °C (battery), 27–37 °C (palm rest), and 40–50 °C (exhaust) published bands.

A three-level grid refinement study gives GCI $\leq$ 1.7 % for the lateral metrics; the SoC is mesh-sensitive due to bounding-box quantization but stays in band.

Future work: (i) real CAD geometry via STEP voxelization; (ii) k–ω SST turbulence in 3D with an algebraic multigrid pressure preconditioner to scale past 10^6 cells; (iii) a transient solver for throttling behaviour; (iv) conjugate radiation for the keyboard deck.

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